

Numerical Programming

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CP#2

Computational Project

Problem 2.1

- ▶ Topic: Modeling with Ordinary Differential Equations
- ▶ Methods/approaches to be used:
 - ▶ Initial value problem for a system of ordinary differential equations
 - ▶ Implicit implicit numerical schemes, for example Euler, diagonally implicit RK, or implicit RK
 - ▶ Newton- Gaus-Seidel method
 - ▶ Fixed point iterations
 - ▶ The methods must be compared on the same real world problem
- ▶ Tasks:
 - ▶ Choose a real world problem that is described by system of ordinary differemntial equation
 - ▶ Post in chat what problem you are going to solve.
 - ▶ Visualize obtained results.
 - ▶ Provide comparison of the two methods.

Computational Project

Deliverables

- ▶ Two working codes
- ▶ A 2-10 page paper explaining your work: model, approach, experiment, conclusions
- ▶ Papers must contain comparison tables and visualisation supporting your conclusions

Important Notice

- ▶ CP assigned 0 points if:
 - ▶ a model problem provided twice by students. Make sure, your model is different from models given by others.
 - ▶ submitted results are not reproducible.
 - ▶ student cannot apply his own code for the input data provided by TA or instructor.
- ▶ Submission deadline: one week after the date of CP's publication.